

# Machine Learning

## Assignment 1 – Polynomial Regression

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### Introduction

The objective of this assignment is to develop polynomial regression models for two datasets and predict the continuous target variable  $y$ . Polynomial regression is used because the relationship between the input features and the target may be nonlinear.

Different polynomial degrees are evaluated using cross-validation to select an appropriate model complexity while avoiding underfitting and overfitting. Mean Squared Error (MSE) is used as the primary model selection metric and  $R^2$  is also calculated to measure the quality of the fit.

### Var 1 – Power Plant Steam Turbine Optimization

#### Approach

The training data is first separated into input features  $X$  and target  $y$ . Polynomial features are generated using `PolynomialFeatures`. The resulting features are standardized using `StandardScaler`.

For each degree from 1 to 10, a hyperparameter search is performed using 5-fold cross-validation with shuffling. Linear Regression and Ridge regression are evaluated across all degrees. Additionally, to better control model complexity and prevent overfitting at lower polynomial levels, Lasso and ElasticNet regressions are evaluated for degrees  $\leq 6$ .

The regularization parameters ( $\alpha$  and `l1_ratio`) are searched over predefined sets of values. The model configuration with the lowest average cross-validation MSE is selected for each degree, and the overall optimal degree is chosen based on the minimum cross-validation MSE across all tested degrees.

#### Optimal Degree

The optimal polynomial degree for `var1` is 5. The following graph shows the cross-validation MSE obtained for the different polynomial degrees.

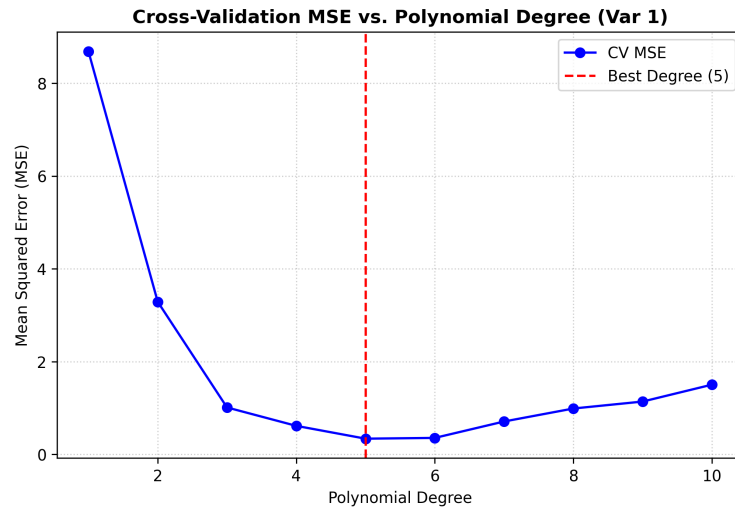


Figure 1: Polynomial degree vs MSE for var1

The degree corresponding to the minimum validation MSE provides the best trade-off between model complexity and prediction error. Increasing the degree beyond this point can introduce unnecessary higher-order and interaction terms and may result in overfitting.

```
=====
Processing Dataset: var1 (1000 rows, 6 features)
=====
Degree 1| Model: lasso      | CV R2 : 0.14116 | CV MSE: 8.6859
Degree 2| Model: lasso      | CV R2 : 0.67494 | CV MSE: 3.2897
Degree 3| Model: elasticnet | CV R2 : 0.89992 | CV MSE: 1.0124
Degree 4| Model: lasso      | CV R2 : 0.93917 | CV MSE: 0.61575
Degree 5| Model: lasso      | CV R2 : 0.96652 | CV MSE: 0.33936
Degree 6| Model: lasso      | CV R2 : 0.96483 | CV MSE: 0.35611
Degree 7| Model: ridge      | CV R2 : 0.92958 | CV MSE: 0.70987
Degree 8| Model: ridge      | CV R2 : 0.90157 | CV MSE: 0.98943
Degree 9| Model: ridge      | CV R2 : 0.88702 | CV MSE: 1.1394
Degree 10| Model: ridge     | CV R2 : 0.85059 | CV MSE: 1.5035

Optimal Configuration Found for var1:
1. Best Degree   : 5
2. Best Model    : LASSO
3. Best Alpha    : 0.01
4. Best L1 Ratio : 1.00
5. Best CV MSE   : 0.33936
6. Best CV R2  : 0.96652
```

Figure 2: Output for var1

## Var 2 – Subterranean Thermal Reservoir Mapping

### Approach

A similar modelling pipeline is used for Var 2, with polynomial features generated for degrees ranging from 1 to 20.

Five-fold shuffled cross-validation is utilized for evaluation. Linear Regression and Ridge regression are tested across all 20 degrees, while Lasso and ElasticNet regressions are additionally incorporated for degrees  $\leq 9$  to handle the higher risk of overfitting introduced by lower-to-moderate degree expansions.

The configuration yielding the lowest average cross-validation MSE determines both the optimal polynomial degree and the corresponding regularization strength.

### Optimal Degree

The optimal polynomial degree for var2 is 10. The following graph shows the cross-validation MSE for the different polynomial degrees.

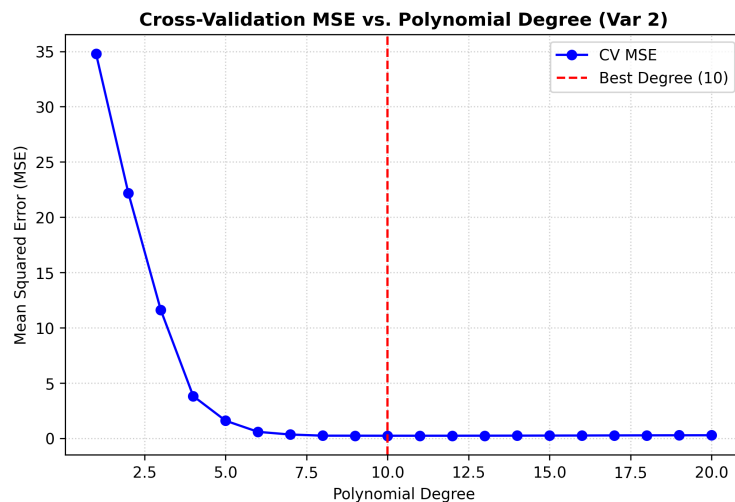


Figure 3: Polynomial degree vs MSE for var2

The minimum of the validation-error curve is used to determine the appropriate model complexity. The use of cross-validation helps prevent choosing a high-degree polynomial simply because it gives a lower training error.

## Output

```
=====
Processing Dataset: var2 (1000 rows, 3 features)
=====
Degree 1| Model: elasticnet | CV R2 : 0.2644 | CV MSE: 34.8
Degree 2| Model: ridge     | CV R2 : 0.53125 | CV MSE: 22.165
Degree 3| Model: ridge     | CV R2 : 0.7532 | CV MSE: 11.593
Degree 4| Model: elasticnet | CV R2 : 0.91871 | CV MSE: 3.814
Degree 5| Model: elasticnet | CV R2 : 0.96602 | CV MSE: 1.5855
Degree 6| Model: ridge     | CV R2 : 0.98765 | CV MSE: 0.57847
Degree 7| Model: ridge     | CV R2 : 0.99275 | CV MSE: 0.33675
Degree 8| Model: elasticnet | CV R2 : 0.995 | CV MSE: 0.23327
Degree 9| Model: ridge     | CV R2 : 0.99507 | CV MSE: 0.23086
Degree 10| Model: ridge    | CV R2 : 0.99513 | CV MSE: 0.22769
Degree 11| Model: ridge    | CV R2 : 0.99513 | CV MSE: 0.22795
Degree 12| Model: ridge    | CV R2 : 0.99512 | CV MSE: 0.2287
Degree 13| Model: ridge    | CV R2 : 0.99506 | CV MSE: 0.2318
Degree 14| Model: ridge    | CV R2 : 0.99497 | CV MSE: 0.23643
Degree 15| Model: ridge    | CV R2 : 0.99485 | CV MSE: 0.24192
Degree 16| Model: ridge    | CV R2 : 0.99473 | CV MSE: 0.24782
Degree 17| Model: ridge    | CV R2 : 0.99455 | CV MSE: 0.25646
Degree 18| Model: ridge    | CV R2 : 0.9944 | CV MSE: 0.26369
Degree 19| Model: ridge    | CV R2 : 0.99428 | CV MSE: 0.26857
Degree 20| Model: ridge    | CV R2 : 0.99423 | CV MSE: 0.27146

Optimal Configuration Found for var2:
1. Best Degree   : 10
2. Best Model    : RIDGE
3. Best Alpha    : 1
4. Best L1 Ratio : 0.00
5. Best CV MSE   : 0.22769
6. Best CV R^2   : 0.99513
```

Figure 4: Output for var2

## Techniques Used

The main techniques used in the implementation are polynomial feature expansion, feature standardization, Linear Regression, Ridge regression, Lasso regression, ElasticNet regression, and 5-fold shuffled cross-validation.

Polynomial feature expansion allows linear regression models to capture nonlinear relationships. Standardization ensures that all generated features are placed on comparable scales prior to regularization.

Ridge regression adds an  $L_2$  penalty to the model coefficients, while Lasso regression applies an  $L_1$  penalty. ElasticNet combines both  $L_1$  and  $L_2$  penalties, controlled by an  $l_1$  ratio parameter, providing a balanced approach for correlated features. These regularization techniques help control the complexity of polynomial models, particularly when higher-degree expansions produce many correlated terms.

The hyperparameter search spaces tested in the implementation are:

$$\alpha_{\text{Ridge}} = \{10^{-8}, 10^{-6}, 10^{-4}, 10^{-3}, 10^{-2}, 10^{-1}, 1, 10\}$$

$$\alpha_{\text{Lasso}} = \{10^{-4}, 10^{-3}, 3 \times 10^{-3}, 10^{-2}, 3 \times 10^{-2}\}$$

$$\alpha_{\text{ElasticNet}} = \{10^{-4}, 10^{-3}, 3 \times 10^{-3}, 10^{-2}, 3 \times 10^{-2}\}$$

and

$$\text{ll\_ratio}_{\text{ElasticNet}} = \{0.1, 0.3, 0.5, 0.7, 0.9\}.$$

## Evaluation Metrics

The primary metric used for selecting the model is Mean Squared Error:

$$MSE = \frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)^2.$$

A lower MSE indicates better prediction performance.

The coefficient of determination is also calculated:

$$R^2 = 1 - \frac{\sum_i (y_i - \hat{y}_i)^2}{\sum_i (y_i - \bar{y})^2}.$$

A higher  $R^2$  indicates that the model explains a larger proportion of the variation in the target variable.

## Final Model and Conclusion

After selecting the optimal polynomial degree and regularization configuration, the selected pipeline is retrained using the complete training dataset. It is then used to generate predictions for the corresponding test dataset.

The optimal degree is degree 5 for Var 1 and degree 10 for Var 2. The model searches the allowed degree ranges and regularization parameters and selects the configuration with the minimum cross-validation MSE.

Overall, polynomial regression provides a suitable approach for both problems because it can model nonlinear relationships while cross-validation and regularization help control model complexity. The final predictions are saved separately for the two test datasets.